

Imports

```
In [1]: import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

import torch
import torch.nn as nn
import torch.optim as optim

from torch.utils.data import Dataset, DataLoader, random_split
```

Configuration

```
In [ ]: TRAIN_CSV = "../data/kaggle/fashion-mnist_train.csv"
TEST_CSV = "../data/kaggle/fashion-mnist_test.csv"

BATCH_SIZE = 64
LEARNING_RATE = 0.001
NUM_EPOCHS = 50
VALID_RATIO = 0.2 # 20% of the training dataset will be used as validation
RANDOM_SEED = 42

CLASS_NAMES = [
    "T-shirt/top",
    "Trouser",
    "Pullover",
    "Dress",
    "Coat",
    "Sandal",
    "Shirt",
    "Sneaker",
    "Bag",
    "Ankle boot"
]

torch.manual_seed(RANDOM_SEED)
np.random.seed(RANDOM_SEED)

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Using device:", device)
```

Using device: cpu

Check files

```
In [3]: if not os.path.exists(TRAIN_CSV):
        raise FileNotFoundError(f"Training file not found: {TRAIN_CSV}")

if not os.path.exists(TEST_CSV):
    raise FileNotFoundError(f"Test file not found: {TEST_CSV}")

print("Both files found.")
```

Both files found.

Create custom dataset

```
In [4]: class FashionMNISTCSVDataset(Dataset):
    def __init__(self, csv_file):
        df = pd.read_csv(csv_file)

        self.labels = df.iloc[:, 0].to_numpy(dtype=np.int64)
        self.images = df.iloc[:, 1:].to_numpy(dtype=np.float32) / 255.0
        self.images = self.images.reshape(-1, 1, 28, 28) # (number_of_ima

    def __len__(self):
        return len(self.labels)

    def __getitem__(self, idx): # get the sample at position idx
        image = torch.from_numpy(self.images[idx])
        label = torch.tensor(self.labels[idx], dtype=torch.long)
        return image, label
```

Load datasets

```
In [5]: full_train_dataset = FashionMNISTCSVDataset(TRAIN_CSV)
        test_dataset = FashionMNISTCSVDataset(TEST_CSV)

        print("Full train dataset size:", len(full_train_dataset))
        print("Test dataset size:", len(test_dataset))
```

Full train dataset size: 60000

Test dataset size: 10000

Split into train and validation

```
In [6]: train_size = int((1 - VALID_RATIO) * len(full_train_dataset))
        val_size = len(full_train_dataset) - train_size

        train_dataset, val_dataset = random_split(
            full_train_dataset,
            [train_size, val_size],
            generator=torch.Generator().manual_seed(RANDOM_SEED)
        )

        print("Train size:", len(train_dataset))
        print("Validation size:", len(val_dataset))
```

Train size: 48000

Validation size: 12000

Create DataLoaders

```
In [7]: train_loader = DataLoader(train_dataset, batch_size=BATCH_SIZE, shuffle=True)
        val_loader = DataLoader(val_dataset, batch_size=BATCH_SIZE, shuffle=False)
        test_loader = DataLoader(test_dataset, batch_size=BATCH_SIZE, shuffle=False)
```

Define the CNN model

```
In [ ]: class FashionCNNFlexible(nn.Module):
    def __init__(
        self,
        conv_channels=[32, 64],
        kernel_size=3,
        pool_type="max",
```

```

activation="relu",
dropout=0.3,
use_batchnorm=False,
num_classes=10
):
    super().__init__()

    layers = []
    in_channels = 1
    current_size = 28

    def get_activation():
        if activation.lower() == "relu":
            return nn.ReLU()
        elif activation.lower() == "gelu":
            return nn.GELU()
        else:
            raise ValueError(f"Unsupported activation: {activation}")

    def get_pool():
        if pool_type.lower() == "max":
            return nn.MaxPool2d(kernel_size=2, stride=2)
        elif pool_type.lower() == "avg":
            return nn.AvgPool2d(kernel_size=2, stride=2)
        else:
            raise ValueError(f"Unsupported pool type: {pool_type}")

    padding = kernel_size // 2 # keep spatial size same after conv

    for out_channels in conv_channels:
        layers.append(nn.Conv2d(in_channels, out_channels, kernel_size=
# (current_size + 2*padding - kernel_size)//stride + 1 = input

        if use_batchnorm:
            layers.append(nn.BatchNorm2d(out_channels))

        layers.append(get_activation())
        layers.append(get_pool())

        in_channels = out_channels
        current_size = current_size // 2 # after 2x2 pooling

    self.features = nn.Sequential(*layers)

    flattened_dim = conv_channels[-1] * current_size * current_size

    self.classifier = nn.Sequential(
        nn.Flatten(),
        nn.Linear(flattened_dim, 128),
        get_activation(),
        nn.Dropout(dropout),
        nn.Linear(128, num_classes)
    )

    def forward(self, x):
        x = self.features(x)
        x = self.classifier(x)
        return x

```

```
In [10]: model_configs = {
    "V1_Baseline": {
        "conv_channels": [32, 64],
        "kernel_size": 3,
        "pool_type": "max",
        "activation": "relu",
        "dropout": 0.3,
        "use_batchnorm": False
    },
    "V2_BatchNorm": {
        "conv_channels": [32, 64],
        "kernel_size": 3,
        "pool_type": "max",
        "activation": "relu",
        "dropout": 0.3,
        "use_batchnorm": True
    },
    "V3_LargerKernel": {
        "conv_channels": [32, 64],
        "kernel_size": 5,
        "pool_type": "max",
        "activation": "relu",
        "dropout": 0.3,
        "use_batchnorm": False
    },
    "V4_AvgPool": {
        "conv_channels": [32, 64],
        "kernel_size": 3,
        "pool_type": "avg",
        "activation": "relu",
        "dropout": 0.3,
        "use_batchnorm": False
    },
    "V5_GELU": {
        "conv_channels": [32, 64],
        "kernel_size": 3,
        "pool_type": "max",
        "activation": "gelu",
        "dropout": 0.3,
        "use_batchnorm": False
    },
    "V6_NoDropout": {
        "conv_channels": [32, 64],
        "kernel_size": 3,
        "pool_type": "max",
        "activation": "relu",
        "dropout": 0.0,
        "use_batchnorm": False
    },
    "V7_Deeper": {
        "conv_channels": [16, 32, 64],
        "kernel_size": 3,
        "pool_type": "max",
        "activation": "relu",
        "dropout": 0.3,
        "use_batchnorm": True
    }
}
```

Initialize model, loss, optimizer

```
In [11]: model = FashionCNNFlexible().to(device)
         criterion = nn.CrossEntropyLoss()
         optimizer = optim.Adam(model.parameters(), lr=LEARNING_RATE)

         print(model)

FashionCNNFlexible(
  (features): Sequential(
    (0): Conv2d(1, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): ReLU()
    (2): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
    (3): Conv2d(32, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (4): ReLU()
    (5): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
  )
  (classifier): Sequential(
    (0): Flatten(start_dim=1, end_dim=-1)
    (1): Linear(in_features=3136, out_features=128, bias=True)
    (2): ReLU()
    (3): Dropout(p=0.3, inplace=False)
    (4): Linear(in_features=128, out_features=10, bias=True)
  )
)
```

Training function

```
In [12]: def train_one_epoch(model, loader, criterion, optimizer, device):
         model.train()

         running_loss = 0.0
         correct = 0
         total = 0

         for images, labels in loader: # images: [batch_size, 1, 28, 28], labels: [batch_size, 1]
             images = images.to(device, non_blocking=True)
             labels = labels.to(device, non_blocking=True)

             optimizer.zero_grad()
             outputs = model(images) # [batch_size, 10]
             loss = criterion(outputs, labels) # computes loss
             loss.backward() # calculates gradients
             optimizer.step() # updates the model parameters based on the calculated gradients

             running_loss += loss.item() * images.size(0)
             preds = outputs.argmax(dim=1) # predicted class is the index of the maximum value
             correct += (preds == labels).sum().item()
             total += labels.size(0)

         return running_loss / total, correct / total
```

Evaluation function

```
In [13]: def evaluate(model, loader, criterion, device):
         model.eval()
```

```

    running_loss = 0.0
    correct = 0
    total = 0
    all_preds = []
    all_labels = []

    with torch.no_grad(): # tells PyTorch not to calculate gradients, whi
        for images, labels in loader:
            images = images.to(device, non_blocking=True)
            labels = labels.to(device, non_blocking=True)

            outputs = model(images)
            loss = criterion(outputs, labels)

            running_loss += loss.item() * images.size(0)
            preds = outputs.argmax(dim=1)

            correct += (preds == labels).sum().item()
            total += labels.size(0)

            all_preds.extend(preds.cpu().numpy()) # This saves all predic
            all_labels.extend(labels.cpu().numpy())

    return running_loss / total, correct / total, np.array(all_preds), np

```

```

In [14]: import torch
import pandas as pd
import os

all_results = []

os.makedirs("saved_models", exist_ok=True)
os.makedirs("results", exist_ok=True)

for model_name, config in model_configs.items():
    print(f"\nTraining {model_name}...")

    model = FashionCNNFlexible(**config).to(device)
    optimizer = torch.optim.Adam(model.parameters(), lr=LEARNING_RATE)

    train_losses = []
    train_accuracies = []
    val_losses = []
    val_accuracies = []

    best_val_acc = 0.0
    best_val_epoch = 0

    best_val_loss = float("inf")
    best_val_loss_epoch = 0

    for epoch in range(NUM_EPOCHS):
        train_loss, train_acc = train_one_epoch(model, train_loader, crit
        val_loss, val_acc, _, _ = evaluate(model, val_loader, criterion,

        train_losses.append(train_loss)
        train_accuracies.append(train_acc)
        val_losses.append(val_loss)
        val_accuracies.append(val_acc)

```

```

        if val_acc > best_val_acc:
            best_val_acc = val_acc
            best_val_epoch = epoch + 1
            torch.save(model.state_dict(), f"saved_models/{model_name}_best_val_acc.pth")

        # save best model by validation loss
        if val_loss < best_val_loss:
            best_val_loss = val_loss
            best_val_loss_epoch = epoch + 1
            torch.save(
                model.state_dict(),
                f"saved_models/{model_name}_best_loss.pth"
            )

    print(f"{model_name} | Epoch [{epoch+1}/{NUM_EPOCHS}]")
    print(f"Train Loss: {train_loss:.4f} | Train Acc: {train_acc:.4f}")
    print(f"Val Loss: {val_loss:.4f} | Val Acc: {val_acc:.4f}")
    print("-" * 50)

    # save per-epoch results for this model
    model_df = pd.DataFrame({
        "model": model_name,
        "epoch": range(1, NUM_EPOCHS + 1),
        "train_loss": train_losses,
        "train_accuracy": train_accuracies,
        "val_loss": val_losses,
        "val_accuracy": val_accuracies
    })
    model_df.to_csv(f"results/{model_name}_training_log.csv", index=False)

    # save summary result
    all_results.append({
        "model": model_name,
        "best_val_accuracy": best_val_acc,
        "best_val_epoch": best_val_epoch,
        "best_val_loss": best_val_loss,
        "best_val_loss_epoch": best_val_loss_epoch,
        "final_train_loss": train_losses[-1],
        "final_train_accuracy": train_accuracies[-1],
        "final_val_loss": val_losses[-1],
        "final_val_accuracy": val_accuracies[-1]
    })

    # save summary of all models
    summary_df = pd.DataFrame(all_results)

    summary_by_acc_df = summary_df.sort_values(by="best_val_accuracy", ascending=False)
    summary_by_acc_df.to_csv("results/model_comparison_summary.csv", index=False)

    summary_by_loss_df = summary_df.sort_values(by="best_val_loss", ascending=True)
    summary_by_loss_df.to_csv("results/model_comparison_summary_by_loss.csv", index=False)

    print("\nAll models completed.")
    print("Sorted by best validation accuracy:")
    print(summary_df)

    print("\nSorted by best validation loss:")
    print(summary_by_loss_df)

```

Training V1_Baseline...

```
/Users/jiacheng/Documents/NUS_Course/CS3244/group_project/cnn_env/lib/python3.10/site-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true but not supported on MPS now, device pinned memory won't be used.  
  super().__init__(loader)
```


V1_Baseline | Epoch [1/50]
Train Loss: 0.5469 | Train Acc: 0.8009
Val Loss: 0.3592 | Val Acc: 0.8672

V1_Baseline | Epoch [2/50]
Train Loss: 0.3456 | Train Acc: 0.8757
Val Loss: 0.2998 | Val Acc: 0.8897

V1_Baseline | Epoch [3/50]
Train Loss: 0.2992 | Train Acc: 0.8931
Val Loss: 0.2684 | Val Acc: 0.9034

V1_Baseline | Epoch [4/50]
Train Loss: 0.2643 | Train Acc: 0.9042
Val Loss: 0.2507 | Val Acc: 0.9094

V1_Baseline | Epoch [5/50]
Train Loss: 0.2391 | Train Acc: 0.9111
Val Loss: 0.2465 | Val Acc: 0.9063

V1_Baseline | Epoch [6/50]
Train Loss: 0.2202 | Train Acc: 0.9182
Val Loss: 0.2316 | Val Acc: 0.9172

V1_Baseline | Epoch [7/50]
Train Loss: 0.2020 | Train Acc: 0.9265
Val Loss: 0.2267 | Val Acc: 0.9177

V1_Baseline | Epoch [8/50]
Train Loss: 0.1852 | Train Acc: 0.9328
Val Loss: 0.2156 | Val Acc: 0.9209

V1_Baseline | Epoch [9/50]
Train Loss: 0.1713 | Train Acc: 0.9359
Val Loss: 0.2234 | Val Acc: 0.9203

V1_Baseline | Epoch [10/50]
Train Loss: 0.1562 | Train Acc: 0.9417
Val Loss: 0.2168 | Val Acc: 0.9232

V1_Baseline | Epoch [11/50]
Train Loss: 0.1447 | Train Acc: 0.9462
Val Loss: 0.2339 | Val Acc: 0.9205

V1_Baseline | Epoch [12/50]
Train Loss: 0.1346 | Train Acc: 0.9493
Val Loss: 0.2363 | Val Acc: 0.9223

V1_Baseline | Epoch [13/50]
Train Loss: 0.1228 | Train Acc: 0.9532
Val Loss: 0.2468 | Val Acc: 0.9202

V1_Baseline | Epoch [14/50]
Train Loss: 0.1123 | Train Acc: 0.9581
Val Loss: 0.2298 | Val Acc: 0.9232

V1_Baseline | Epoch [15/50]
Train Loss: 0.1044 | Train Acc: 0.9609
Val Loss: 0.2294 | Val Acc: 0.9244

V1_Baseline | Epoch [16/50]
Train Loss: 0.0978 | Train Acc: 0.9632
Val Loss: 0.2407 | Val Acc: 0.9199

V1_Baseline | Epoch [17/50]
Train Loss: 0.0893 | Train Acc: 0.9660
Val Loss: 0.2582 | Val Acc: 0.9240

V1_Baseline | Epoch [18/50]
Train Loss: 0.0832 | Train Acc: 0.9686
Val Loss: 0.2569 | Val Acc: 0.9235

V1_Baseline | Epoch [19/50]
Train Loss: 0.0789 | Train Acc: 0.9696
Val Loss: 0.2758 | Val Acc: 0.9265

V1_Baseline | Epoch [20/50]
Train Loss: 0.0722 | Train Acc: 0.9722
Val Loss: 0.2838 | Val Acc: 0.9229

V1_Baseline | Epoch [21/50]
Train Loss: 0.0673 | Train Acc: 0.9739
Val Loss: 0.3115 | Val Acc: 0.9203

V1_Baseline | Epoch [22/50]
Train Loss: 0.0652 | Train Acc: 0.9747
Val Loss: 0.3053 | Val Acc: 0.9269

V1_Baseline | Epoch [23/50]
Train Loss: 0.0608 | Train Acc: 0.9761
Val Loss: 0.3121 | Val Acc: 0.9243

V1_Baseline | Epoch [24/50]
Train Loss: 0.0577 | Train Acc: 0.9778
Val Loss: 0.3470 | Val Acc: 0.9209

V1_Baseline | Epoch [25/50]
Train Loss: 0.0555 | Train Acc: 0.9789
Val Loss: 0.3350 | Val Acc: 0.9243

V1_Baseline | Epoch [26/50]
Train Loss: 0.0561 | Train Acc: 0.9785
Val Loss: 0.3314 | Val Acc: 0.9233

V1_Baseline | Epoch [27/50]
Train Loss: 0.0496 | Train Acc: 0.9808
Val Loss: 0.3410 | Val Acc: 0.9267

V1_Baseline | Epoch [28/50]
Train Loss: 0.0499 | Train Acc: 0.9805
Val Loss: 0.3627 | Val Acc: 0.9263

V1_Baseline | Epoch [29/50]
Train Loss: 0.0476 | Train Acc: 0.9817
Val Loss: 0.3390 | Val Acc: 0.9252

V1_Baseline | Epoch [30/50]
Train Loss: 0.0463 | Train Acc: 0.9824
Val Loss: 0.3765 | Val Acc: 0.9235

V1_Baseline | Epoch [31/50]
Train Loss: 0.0452 | Train Acc: 0.9829
Val Loss: 0.3732 | Val Acc: 0.9243

V1_Baseline | Epoch [32/50]
Train Loss: 0.0448 | Train Acc: 0.9826
Val Loss: 0.4228 | Val Acc: 0.9224

V1_Baseline | Epoch [33/50]
Train Loss: 0.0409 | Train Acc: 0.9836
Val Loss: 0.3843 | Val Acc: 0.9261

V1_Baseline | Epoch [34/50]
Train Loss: 0.0434 | Train Acc: 0.9834
Val Loss: 0.3978 | Val Acc: 0.9245

V1_Baseline | Epoch [35/50]
Train Loss: 0.0380 | Train Acc: 0.9856
Val Loss: 0.3937 | Val Acc: 0.9218

V1_Baseline | Epoch [36/50]
Train Loss: 0.0386 | Train Acc: 0.9847
Val Loss: 0.4057 | Val Acc: 0.9237

V1_Baseline | Epoch [37/50]
Train Loss: 0.0376 | Train Acc: 0.9854
Val Loss: 0.4442 | Val Acc: 0.9197

V1_Baseline | Epoch [38/50]
Train Loss: 0.0359 | Train Acc: 0.9865
Val Loss: 0.4413 | Val Acc: 0.9211

V1_Baseline | Epoch [39/50]
Train Loss: 0.0389 | Train Acc: 0.9852
Val Loss: 0.4144 | Val Acc: 0.9240

V1_Baseline | Epoch [40/50]
Train Loss: 0.0332 | Train Acc: 0.9871
Val Loss: 0.4730 | Val Acc: 0.9225

V1_Baseline | Epoch [41/50]
Train Loss: 0.0345 | Train Acc: 0.9869
Val Loss: 0.4607 | Val Acc: 0.9219

V1_Baseline | Epoch [42/50]
Train Loss: 0.0357 | Train Acc: 0.9865
Val Loss: 0.4842 | Val Acc: 0.9208

V1_Baseline | Epoch [43/50]
Train Loss: 0.0334 | Train Acc: 0.9875
Val Loss: 0.4684 | Val Acc: 0.9247

V1_Baseline | Epoch [44/50]
Train Loss: 0.0329 | Train Acc: 0.9874
Val Loss: 0.4381 | Val Acc: 0.9252

V1_Baseline | Epoch [45/50]
Train Loss: 0.0353 | Train Acc: 0.9869
Val Loss: 0.4545 | Val Acc: 0.9217

V1_Baseline | Epoch [46/50]
Train Loss: 0.0317 | Train Acc: 0.9878
Val Loss: 0.4497 | Val Acc: 0.9241

V1_Baseline | Epoch [47/50]
Train Loss: 0.0332 | Train Acc: 0.9879
Val Loss: 0.4967 | Val Acc: 0.9203

V1_Baseline | Epoch [48/50]
Train Loss: 0.0294 | Train Acc: 0.9883
Val Loss: 0.4857 | Val Acc: 0.9246

V1_Baseline | Epoch [49/50]
Train Loss: 0.0285 | Train Acc: 0.9892
Val Loss: 0.5552 | Val Acc: 0.9221

V1_Baseline | Epoch [50/50]
Train Loss: 0.0308 | Train Acc: 0.9879
Val Loss: 0.5348 | Val Acc: 0.9201

Training V2_BatchNorm...

V2_BatchNorm | Epoch [1/50]
Train Loss: 0.4493 | Train Acc: 0.8371
Val Loss: 0.3135 | Val Acc: 0.8851

V2_BatchNorm | Epoch [2/50]
Train Loss: 0.3139 | Train Acc: 0.8864
Val Loss: 0.2851 | Val Acc: 0.8945

V2_BatchNorm | Epoch [3/50]
Train Loss: 0.2674 | Train Acc: 0.9025
Val Loss: 0.2491 | Val Acc: 0.9086

V2_BatchNorm | Epoch [4/50]
Train Loss: 0.2411 | Train Acc: 0.9110
Val Loss: 0.2674 | Val Acc: 0.9021

V2_BatchNorm | Epoch [5/50]
Train Loss: 0.2224 | Train Acc: 0.9174
Val Loss: 0.2411 | Val Acc: 0.9112

V2_BatchNorm | Epoch [6/50]
Train Loss: 0.1991 | Train Acc: 0.9266
Val Loss: 0.2248 | Val Acc: 0.9191

V2_BatchNorm | Epoch [7/50]
Train Loss: 0.1829 | Train Acc: 0.9324
Val Loss: 0.2226 | Val Acc: 0.9204

V2_BatchNorm | Epoch [8/50]
Train Loss: 0.1676 | Train Acc: 0.9374
Val Loss: 0.2150 | Val Acc: 0.9244

V2_BatchNorm | Epoch [9/50]
Train Loss: 0.1544 | Train Acc: 0.9424
Val Loss: 0.2312 | Val Acc: 0.9193

V2_BatchNorm | Epoch [10/50]
Train Loss: 0.1422 | Train Acc: 0.9467

Val Loss: 0.2411 | Val Acc: 0.9207

V2_BatchNorm | Epoch [11/50]

Train Loss: 0.1295 | Train Acc: 0.9510

Val Loss: 0.2455 | Val Acc: 0.9208

V2_BatchNorm | Epoch [12/50]

Train Loss: 0.1197 | Train Acc: 0.9558

Val Loss: 0.2332 | Val Acc: 0.9253

V2_BatchNorm | Epoch [13/50]

Train Loss: 0.1089 | Train Acc: 0.9586

Val Loss: 0.2573 | Val Acc: 0.9225

V2_BatchNorm | Epoch [14/50]

Train Loss: 0.1016 | Train Acc: 0.9618

Val Loss: 0.2544 | Val Acc: 0.9240

V2_BatchNorm | Epoch [15/50]

Train Loss: 0.0937 | Train Acc: 0.9648

Val Loss: 0.2475 | Val Acc: 0.9221

V2_BatchNorm | Epoch [16/50]

Train Loss: 0.0873 | Train Acc: 0.9662

Val Loss: 0.2633 | Val Acc: 0.9232

V2_BatchNorm | Epoch [17/50]

Train Loss: 0.0796 | Train Acc: 0.9699

Val Loss: 0.2701 | Val Acc: 0.9261

V2_BatchNorm | Epoch [18/50]

Train Loss: 0.0754 | Train Acc: 0.9711

Val Loss: 0.2953 | Val Acc: 0.9236

V2_BatchNorm | Epoch [19/50]

Train Loss: 0.0696 | Train Acc: 0.9736

Val Loss: 0.2996 | Val Acc: 0.9197

V2_BatchNorm | Epoch [20/50]

Train Loss: 0.0704 | Train Acc: 0.9732

Val Loss: 0.3012 | Val Acc: 0.9234

V2_BatchNorm | Epoch [21/50]

Train Loss: 0.0608 | Train Acc: 0.9768

Val Loss: 0.3181 | Val Acc: 0.9187

V2_BatchNorm | Epoch [22/50]

Train Loss: 0.0600 | Train Acc: 0.9774

Val Loss: 0.3094 | Val Acc: 0.9171

V2_BatchNorm | Epoch [23/50]

Train Loss: 0.0589 | Train Acc: 0.9770

Val Loss: 0.3494 | Val Acc: 0.9185

V2_BatchNorm | Epoch [24/50]

Train Loss: 0.0566 | Train Acc: 0.9785

Val Loss: 0.3460 | Val Acc: 0.9216

V2_BatchNorm | Epoch [25/50]

Train Loss: 0.0539 | Train Acc: 0.9802

Val Loss: 0.3306 | Val Acc: 0.9229

V2_BatchNorm | Epoch [26/50]

Train Loss: 0.0541 | Train Acc: 0.9800

Val Loss: 0.3748 | Val Acc: 0.9216

V2_BatchNorm | Epoch [27/50]

Train Loss: 0.0490 | Train Acc: 0.9812

Val Loss: 0.3509 | Val Acc: 0.9238

V2_BatchNorm | Epoch [28/50]

Train Loss: 0.0444 | Train Acc: 0.9827

Val Loss: 0.3767 | Val Acc: 0.9208

V2_BatchNorm | Epoch [29/50]

Train Loss: 0.0476 | Train Acc: 0.9823

Val Loss: 0.3630 | Val Acc: 0.9231

V2_BatchNorm | Epoch [30/50]

Train Loss: 0.0427 | Train Acc: 0.9843

Val Loss: 0.4116 | Val Acc: 0.9211

V2_BatchNorm | Epoch [31/50]

Train Loss: 0.0441 | Train Acc: 0.9829

Val Loss: 0.4078 | Val Acc: 0.9205

V2_BatchNorm | Epoch [32/50]

Train Loss: 0.0418 | Train Acc: 0.9841

Val Loss: 0.3751 | Val Acc: 0.9208

V2_BatchNorm | Epoch [33/50]

Train Loss: 0.0411 | Train Acc: 0.9842

Val Loss: 0.3959 | Val Acc: 0.9185

V2_BatchNorm | Epoch [34/50]

Train Loss: 0.0374 | Train Acc: 0.9862

Val Loss: 0.4042 | Val Acc: 0.9237

V2_BatchNorm | Epoch [35/50]

Train Loss: 0.0368 | Train Acc: 0.9866

Val Loss: 0.4350 | Val Acc: 0.9218

V2_BatchNorm | Epoch [36/50]

Train Loss: 0.0376 | Train Acc: 0.9853

Val Loss: 0.4410 | Val Acc: 0.9242

V2_BatchNorm | Epoch [37/50]

Train Loss: 0.0351 | Train Acc: 0.9872

Val Loss: 0.3964 | Val Acc: 0.9231

V2_BatchNorm | Epoch [38/50]

Train Loss: 0.0345 | Train Acc: 0.9870

Val Loss: 0.4294 | Val Acc: 0.9207

V2_BatchNorm | Epoch [39/50]

Train Loss: 0.0377 | Train Acc: 0.9854

Val Loss: 0.4494 | Val Acc: 0.9210

V2_BatchNorm | Epoch [40/50]

Train Loss: 0.0336 | Train Acc: 0.9872

Val Loss: 0.4396 | Val Acc: 0.9223

V2_BatchNorm | Epoch [41/50]

Train Loss: 0.0332 | Train Acc: 0.9879

Val Loss: 0.4352 | Val Acc: 0.9208

V2_BatchNorm | Epoch [42/50]

Train Loss: 0.0326 | Train Acc: 0.9873

Val Loss: 0.4539 | Val Acc: 0.9184

V2_BatchNorm | Epoch [43/50]

Train Loss: 0.0348 | Train Acc: 0.9873

Val Loss: 0.4347 | Val Acc: 0.9206

V2_BatchNorm | Epoch [44/50]

Train Loss: 0.0316 | Train Acc: 0.9876

Val Loss: 0.4602 | Val Acc: 0.9231

V2_BatchNorm | Epoch [45/50]

Train Loss: 0.0321 | Train Acc: 0.9877

Val Loss: 0.4416 | Val Acc: 0.9222

V2_BatchNorm | Epoch [46/50]

Train Loss: 0.0307 | Train Acc: 0.9883

Val Loss: 0.4567 | Val Acc: 0.9203

V2_BatchNorm | Epoch [47/50]

Train Loss: 0.0292 | Train Acc: 0.9890

Val Loss: 0.4963 | Val Acc: 0.9167

V2_BatchNorm | Epoch [48/50]

Train Loss: 0.0304 | Train Acc: 0.9891

Val Loss: 0.4989 | Val Acc: 0.9205

V2_BatchNorm | Epoch [49/50]

Train Loss: 0.0296 | Train Acc: 0.9889

Val Loss: 0.5061 | Val Acc: 0.9192

V2_BatchNorm | Epoch [50/50]

Train Loss: 0.0272 | Train Acc: 0.9901

Val Loss: 0.4989 | Val Acc: 0.9229

Training V3_LargerKernel...

V3_LargerKernel | Epoch [1/50]

Train Loss: 0.5528 | Train Acc: 0.7975

Val Loss: 0.3511 | Val Acc: 0.8727

V3_LargerKernel | Epoch [2/50]

Train Loss: 0.3501 | Train Acc: 0.8738

Val Loss: 0.3061 | Val Acc: 0.8820

V3_LargerKernel | Epoch [3/50]

Train Loss: 0.2973 | Train Acc: 0.8925

Val Loss: 0.2661 | Val Acc: 0.8999

V3_LargerKernel | Epoch [4/50]

Train Loss: 0.2640 | Train Acc: 0.9050

Val Loss: 0.2814 | Val Acc: 0.8948

V3_LargerKernel | Epoch [5/50]
Train Loss: 0.2392 | Train Acc: 0.9116
Val Loss: 0.2359 | Val Acc: 0.9118

V3_LargerKernel | Epoch [6/50]
Train Loss: 0.2156 | Train Acc: 0.9206
Val Loss: 0.2329 | Val Acc: 0.9132

V3_LargerKernel | Epoch [7/50]
Train Loss: 0.2004 | Train Acc: 0.9262
Val Loss: 0.2325 | Val Acc: 0.9144

V3_LargerKernel | Epoch [8/50]
Train Loss: 0.1863 | Train Acc: 0.9307
Val Loss: 0.2318 | Val Acc: 0.9167

V3_LargerKernel | Epoch [9/50]
Train Loss: 0.1708 | Train Acc: 0.9355
Val Loss: 0.2260 | Val Acc: 0.9195

V3_LargerKernel | Epoch [10/50]
Train Loss: 0.1557 | Train Acc: 0.9415
Val Loss: 0.2268 | Val Acc: 0.9187

V3_LargerKernel | Epoch [11/50]
Train Loss: 0.1422 | Train Acc: 0.9464
Val Loss: 0.2307 | Val Acc: 0.9187

V3_LargerKernel | Epoch [12/50]
Train Loss: 0.1305 | Train Acc: 0.9505
Val Loss: 0.2398 | Val Acc: 0.9209

V3_LargerKernel | Epoch [13/50]
Train Loss: 0.1215 | Train Acc: 0.9536
Val Loss: 0.2351 | Val Acc: 0.9226

V3_LargerKernel | Epoch [14/50]
Train Loss: 0.1103 | Train Acc: 0.9568
Val Loss: 0.2543 | Val Acc: 0.9228

V3_LargerKernel | Epoch [15/50]
Train Loss: 0.1016 | Train Acc: 0.9602
Val Loss: 0.2587 | Val Acc: 0.9207

V3_LargerKernel | Epoch [16/50]
Train Loss: 0.0953 | Train Acc: 0.9633
Val Loss: 0.2810 | Val Acc: 0.9204

V3_LargerKernel | Epoch [17/50]
Train Loss: 0.0906 | Train Acc: 0.9642
Val Loss: 0.2936 | Val Acc: 0.9213

V3_LargerKernel | Epoch [18/50]
Train Loss: 0.0849 | Train Acc: 0.9662
Val Loss: 0.2893 | Val Acc: 0.9205

V3_LargerKernel | Epoch [19/50]
Train Loss: 0.0804 | Train Acc: 0.9684
Val Loss: 0.3042 | Val Acc: 0.9221

V3_LargerKernel | Epoch [20/50]
Train Loss: 0.0728 | Train Acc: 0.9716
Val Loss: 0.3168 | Val Acc: 0.9237

V3_LargerKernel | Epoch [21/50]
Train Loss: 0.0715 | Train Acc: 0.9715
Val Loss: 0.3284 | Val Acc: 0.9177

V3_LargerKernel | Epoch [22/50]
Train Loss: 0.0671 | Train Acc: 0.9733
Val Loss: 0.3291 | Val Acc: 0.9237

V3_LargerKernel | Epoch [23/50]
Train Loss: 0.0637 | Train Acc: 0.9749
Val Loss: 0.3575 | Val Acc: 0.9189

V3_LargerKernel | Epoch [24/50]
Train Loss: 0.0626 | Train Acc: 0.9753
Val Loss: 0.3542 | Val Acc: 0.9209

V3_LargerKernel | Epoch [25/50]
Train Loss: 0.0593 | Train Acc: 0.9765
Val Loss: 0.3649 | Val Acc: 0.9234

V3_LargerKernel | Epoch [26/50]
Train Loss: 0.0551 | Train Acc: 0.9785
Val Loss: 0.3775 | Val Acc: 0.9228

V3_LargerKernel | Epoch [27/50]
Train Loss: 0.0551 | Train Acc: 0.9790
Val Loss: 0.4016 | Val Acc: 0.9180

V3_LargerKernel | Epoch [28/50]
Train Loss: 0.0506 | Train Acc: 0.9804
Val Loss: 0.4244 | Val Acc: 0.9200

V3_LargerKernel | Epoch [29/50]
Train Loss: 0.0519 | Train Acc: 0.9804
Val Loss: 0.4216 | Val Acc: 0.9208

V3_LargerKernel | Epoch [30/50]
Train Loss: 0.0476 | Train Acc: 0.9813
Val Loss: 0.4300 | Val Acc: 0.9229

V3_LargerKernel | Epoch [31/50]
Train Loss: 0.0479 | Train Acc: 0.9821
Val Loss: 0.4297 | Val Acc: 0.9208

V3_LargerKernel | Epoch [32/50]
Train Loss: 0.0447 | Train Acc: 0.9828
Val Loss: 0.4192 | Val Acc: 0.9237

V3_LargerKernel | Epoch [33/50]
Train Loss: 0.0477 | Train Acc: 0.9818
Val Loss: 0.4234 | Val Acc: 0.9182

V3_LargerKernel | Epoch [34/50]
Train Loss: 0.0441 | Train Acc: 0.9830
Val Loss: 0.4342 | Val Acc: 0.9220

V3_LargerKernel | Epoch [35/50]
Train Loss: 0.0410 | Train Acc: 0.9841
Val Loss: 0.4897 | Val Acc: 0.9224

V3_LargerKernel | Epoch [36/50]
Train Loss: 0.0418 | Train Acc: 0.9840
Val Loss: 0.4569 | Val Acc: 0.9243

V3_LargerKernel | Epoch [37/50]
Train Loss: 0.0405 | Train Acc: 0.9845
Val Loss: 0.4740 | Val Acc: 0.9204

V3_LargerKernel | Epoch [38/50]
Train Loss: 0.0421 | Train Acc: 0.9840
Val Loss: 0.4523 | Val Acc: 0.9241

V3_LargerKernel | Epoch [39/50]
Train Loss: 0.0389 | Train Acc: 0.9858
Val Loss: 0.4757 | Val Acc: 0.9203

V3_LargerKernel | Epoch [40/50]
Train Loss: 0.0402 | Train Acc: 0.9848
Val Loss: 0.4954 | Val Acc: 0.9229

V3_LargerKernel | Epoch [41/50]
Train Loss: 0.0387 | Train Acc: 0.9853
Val Loss: 0.5288 | Val Acc: 0.9235

V3_LargerKernel | Epoch [42/50]
Train Loss: 0.0357 | Train Acc: 0.9858
Val Loss: 0.4691 | Val Acc: 0.9241

V3_LargerKernel | Epoch [43/50]
Train Loss: 0.0373 | Train Acc: 0.9856
Val Loss: 0.4872 | Val Acc: 0.9204

V3_LargerKernel | Epoch [44/50]
Train Loss: 0.0383 | Train Acc: 0.9856
Val Loss: 0.4973 | Val Acc: 0.9214

V3_LargerKernel | Epoch [45/50]
Train Loss: 0.0357 | Train Acc: 0.9869
Val Loss: 0.5038 | Val Acc: 0.9217

V3_LargerKernel | Epoch [46/50]
Train Loss: 0.0323 | Train Acc: 0.9872
Val Loss: 0.5347 | Val Acc: 0.9245

V3_LargerKernel | Epoch [47/50]
Train Loss: 0.0348 | Train Acc: 0.9867
Val Loss: 0.5141 | Val Acc: 0.9223

V3_LargerKernel | Epoch [48/50]
Train Loss: 0.0386 | Train Acc: 0.9856
Val Loss: 0.5056 | Val Acc: 0.9243

V3_LargerKernel | Epoch [49/50]
Train Loss: 0.0330 | Train Acc: 0.9875
Val Loss: 0.5062 | Val Acc: 0.9234

V3_LargerKernel | Epoch [50/50]
Train Loss: 0.0323 | Train Acc: 0.9880
Val Loss: 0.5651 | Val Acc: 0.9209

Training V4_AvgPool...
V4_AvgPool | Epoch [1/50]
Train Loss: 0.6336 | Train Acc: 0.7644
Val Loss: 0.4395 | Val Acc: 0.8335

V4_AvgPool | Epoch [2/50]
Train Loss: 0.4234 | Train Acc: 0.8445
Val Loss: 0.3658 | Val Acc: 0.8618

V4_AvgPool | Epoch [3/50]
Train Loss: 0.3708 | Train Acc: 0.8653
Val Loss: 0.3311 | Val Acc: 0.8773

V4_AvgPool | Epoch [4/50]
Train Loss: 0.3358 | Train Acc: 0.8776
Val Loss: 0.3185 | Val Acc: 0.8808

V4_AvgPool | Epoch [5/50]
Train Loss: 0.3145 | Train Acc: 0.8876
Val Loss: 0.2946 | Val Acc: 0.8947

V4_AvgPool | Epoch [6/50]
Train Loss: 0.2945 | Train Acc: 0.8934
Val Loss: 0.2741 | Val Acc: 0.8994

V4_AvgPool | Epoch [7/50]
Train Loss: 0.2786 | Train Acc: 0.8981
Val Loss: 0.2728 | Val Acc: 0.9013

V4_AvgPool | Epoch [8/50]
Train Loss: 0.2659 | Train Acc: 0.9047
Val Loss: 0.2687 | Val Acc: 0.9020

V4_AvgPool | Epoch [9/50]
Train Loss: 0.2540 | Train Acc: 0.9080
Val Loss: 0.2573 | Val Acc: 0.9057

V4_AvgPool | Epoch [10/50]
Train Loss: 0.2420 | Train Acc: 0.9118
Val Loss: 0.2481 | Val Acc: 0.9104

V4_AvgPool | Epoch [11/50]
Train Loss: 0.2320 | Train Acc: 0.9165
Val Loss: 0.2474 | Val Acc: 0.9111

V4_AvgPool | Epoch [12/50]
Train Loss: 0.2216 | Train Acc: 0.9175
Val Loss: 0.2510 | Val Acc: 0.9099

V4_AvgPool | Epoch [13/50]
Train Loss: 0.2098 | Train Acc: 0.9227
Val Loss: 0.2383 | Val Acc: 0.9173

V4_AvgPool | Epoch [14/50]
Train Loss: 0.2012 | Train Acc: 0.9253

Val Loss: 0.2436 | Val Acc: 0.9162

V4_AvgPool | Epoch [15/50]

Train Loss: 0.1954 | Train Acc: 0.9280

Val Loss: 0.2433 | Val Acc: 0.9182

V4_AvgPool | Epoch [16/50]

Train Loss: 0.1865 | Train Acc: 0.9300

Val Loss: 0.2355 | Val Acc: 0.9192

V4_AvgPool | Epoch [17/50]

Train Loss: 0.1797 | Train Acc: 0.9322

Val Loss: 0.2408 | Val Acc: 0.9170

V4_AvgPool | Epoch [18/50]

Train Loss: 0.1713 | Train Acc: 0.9364

Val Loss: 0.2387 | Val Acc: 0.9187

V4_AvgPool | Epoch [19/50]

Train Loss: 0.1639 | Train Acc: 0.9384

Val Loss: 0.2369 | Val Acc: 0.9196

V4_AvgPool | Epoch [20/50]

Train Loss: 0.1608 | Train Acc: 0.9392

Val Loss: 0.2438 | Val Acc: 0.9210

V4_AvgPool | Epoch [21/50]

Train Loss: 0.1506 | Train Acc: 0.9440

Val Loss: 0.2485 | Val Acc: 0.9194

V4_AvgPool | Epoch [22/50]

Train Loss: 0.1454 | Train Acc: 0.9459

Val Loss: 0.2539 | Val Acc: 0.9178

V4_AvgPool | Epoch [23/50]

Train Loss: 0.1385 | Train Acc: 0.9472

Val Loss: 0.2523 | Val Acc: 0.9231

V4_AvgPool | Epoch [24/50]

Train Loss: 0.1346 | Train Acc: 0.9487

Val Loss: 0.2464 | Val Acc: 0.9200

V4_AvgPool | Epoch [25/50]

Train Loss: 0.1293 | Train Acc: 0.9506

Val Loss: 0.2462 | Val Acc: 0.9227

V4_AvgPool | Epoch [26/50]

Train Loss: 0.1225 | Train Acc: 0.9538

Val Loss: 0.2637 | Val Acc: 0.9228

V4_AvgPool | Epoch [27/50]

Train Loss: 0.1210 | Train Acc: 0.9544

Val Loss: 0.2719 | Val Acc: 0.9231

V4_AvgPool | Epoch [28/50]

Train Loss: 0.1150 | Train Acc: 0.9565

Val Loss: 0.2553 | Val Acc: 0.9232

V4_AvgPool | Epoch [29/50]

Train Loss: 0.1094 | Train Acc: 0.9580

Val Loss: 0.2646 | Val Acc: 0.9221

V4_AvgPool | Epoch [30/50]

Train Loss: 0.1080 | Train Acc: 0.9585

Val Loss: 0.2634 | Val Acc: 0.9219

V4_AvgPool | Epoch [31/50]

Train Loss: 0.1020 | Train Acc: 0.9606

Val Loss: 0.2703 | Val Acc: 0.9220

V4_AvgPool | Epoch [32/50]

Train Loss: 0.0980 | Train Acc: 0.9624

Val Loss: 0.2877 | Val Acc: 0.9235

V4_AvgPool | Epoch [33/50]

Train Loss: 0.0940 | Train Acc: 0.9630

Val Loss: 0.2878 | Val Acc: 0.9213

V4_AvgPool | Epoch [34/50]

Train Loss: 0.0911 | Train Acc: 0.9636

Val Loss: 0.2871 | Val Acc: 0.9236

V4_AvgPool | Epoch [35/50]

Train Loss: 0.0905 | Train Acc: 0.9657

Val Loss: 0.3046 | Val Acc: 0.9207

V4_AvgPool | Epoch [36/50]

Train Loss: 0.0872 | Train Acc: 0.9664

Val Loss: 0.3150 | Val Acc: 0.9217

V4_AvgPool | Epoch [37/50]

Train Loss: 0.0823 | Train Acc: 0.9671

Val Loss: 0.3248 | Val Acc: 0.9233

V4_AvgPool | Epoch [38/50]

Train Loss: 0.0823 | Train Acc: 0.9680

Val Loss: 0.3100 | Val Acc: 0.9221

V4_AvgPool | Epoch [39/50]

Train Loss: 0.0770 | Train Acc: 0.9701

Val Loss: 0.3230 | Val Acc: 0.9244

V4_AvgPool | Epoch [40/50]

Train Loss: 0.0756 | Train Acc: 0.9703

Val Loss: 0.3273 | Val Acc: 0.9217

V4_AvgPool | Epoch [41/50]

Train Loss: 0.0726 | Train Acc: 0.9713

Val Loss: 0.3273 | Val Acc: 0.9224

V4_AvgPool | Epoch [42/50]

Train Loss: 0.0714 | Train Acc: 0.9730

Val Loss: 0.3484 | Val Acc: 0.9227

V4_AvgPool | Epoch [43/50]

Train Loss: 0.0691 | Train Acc: 0.9726

Val Loss: 0.3569 | Val Acc: 0.9224

V4_AvgPool | Epoch [44/50]

Train Loss: 0.0693 | Train Acc: 0.9724

Val Loss: 0.3614 | Val Acc: 0.9195

V4_AvgPool | Epoch [45/50]

Train Loss: 0.0662 | Train Acc: 0.9737

Val Loss: 0.3331 | Val Acc: 0.9246

V4_AvgPool | Epoch [46/50]

Train Loss: 0.0620 | Train Acc: 0.9753

Val Loss: 0.3566 | Val Acc: 0.9234

V4_AvgPool | Epoch [47/50]

Train Loss: 0.0642 | Train Acc: 0.9751

Val Loss: 0.3759 | Val Acc: 0.9248

V4_AvgPool | Epoch [48/50]

Train Loss: 0.0599 | Train Acc: 0.9763

Val Loss: 0.3747 | Val Acc: 0.9237

V4_AvgPool | Epoch [49/50]

Train Loss: 0.0592 | Train Acc: 0.9765

Val Loss: 0.3681 | Val Acc: 0.9250

V4_AvgPool | Epoch [50/50]

Train Loss: 0.0572 | Train Acc: 0.9773

Val Loss: 0.3769 | Val Acc: 0.9263

Training V5_GELU...

V5_GELU | Epoch [1/50]

Train Loss: 0.5073 | Train Acc: 0.8135

Val Loss: 0.3489 | Val Acc: 0.8759

V5_GELU | Epoch [2/50]

Train Loss: 0.3308 | Train Acc: 0.8792

Val Loss: 0.2961 | Val Acc: 0.8932

V5_GELU | Epoch [3/50]

Train Loss: 0.2821 | Train Acc: 0.8966

Val Loss: 0.2699 | Val Acc: 0.9002

V5_GELU | Epoch [4/50]

Train Loss: 0.2488 | Train Acc: 0.9074

Val Loss: 0.2521 | Val Acc: 0.9103

V5_GELU | Epoch [5/50]

Train Loss: 0.2224 | Train Acc: 0.9165

Val Loss: 0.2358 | Val Acc: 0.9128

V5_GELU | Epoch [6/50]

Train Loss: 0.1994 | Train Acc: 0.9255

Val Loss: 0.2331 | Val Acc: 0.9162

V5_GELU | Epoch [7/50]

Train Loss: 0.1816 | Train Acc: 0.9317

Val Loss: 0.2381 | Val Acc: 0.9162

V5_GELU | Epoch [8/50]

Train Loss: 0.1597 | Train Acc: 0.9396

Val Loss: 0.2423 | Val Acc: 0.9179

V5_GELU | Epoch [9/50]
Train Loss: 0.1431 | Train Acc: 0.9464
Val Loss: 0.2355 | Val Acc: 0.9207

V5_GELU | Epoch [10/50]
Train Loss: 0.1291 | Train Acc: 0.9523
Val Loss: 0.2421 | Val Acc: 0.9224

V5_GELU | Epoch [11/50]
Train Loss: 0.1154 | Train Acc: 0.9572
Val Loss: 0.2406 | Val Acc: 0.9261

V5_GELU | Epoch [12/50]
Train Loss: 0.1041 | Train Acc: 0.9610
Val Loss: 0.2474 | Val Acc: 0.9261

V5_GELU | Epoch [13/50]
Train Loss: 0.0899 | Train Acc: 0.9664
Val Loss: 0.2674 | Val Acc: 0.9211

V5_GELU | Epoch [14/50]
Train Loss: 0.0811 | Train Acc: 0.9701
Val Loss: 0.2790 | Val Acc: 0.9235

V5_GELU | Epoch [15/50]
Train Loss: 0.0707 | Train Acc: 0.9737
Val Loss: 0.2733 | Val Acc: 0.9241

V5_GELU | Epoch [16/50]
Train Loss: 0.0641 | Train Acc: 0.9758
Val Loss: 0.2784 | Val Acc: 0.9252

V5_GELU | Epoch [17/50]
Train Loss: 0.0556 | Train Acc: 0.9799
Val Loss: 0.3040 | Val Acc: 0.9235

V5_GELU | Epoch [18/50]
Train Loss: 0.0538 | Train Acc: 0.9795
Val Loss: 0.3170 | Val Acc: 0.9233

V5_GELU | Epoch [19/50]
Train Loss: 0.0505 | Train Acc: 0.9812
Val Loss: 0.3149 | Val Acc: 0.9263

V5_GELU | Epoch [20/50]
Train Loss: 0.0434 | Train Acc: 0.9835
Val Loss: 0.3296 | Val Acc: 0.9239

V5_GELU | Epoch [21/50]
Train Loss: 0.0425 | Train Acc: 0.9842
Val Loss: 0.3363 | Val Acc: 0.9217

V5_GELU | Epoch [22/50]
Train Loss: 0.0385 | Train Acc: 0.9858
Val Loss: 0.3523 | Val Acc: 0.9237

V5_GELU | Epoch [23/50]
Train Loss: 0.0373 | Train Acc: 0.9865
Val Loss: 0.3585 | Val Acc: 0.9227

V5_GELU | Epoch [24/50]
Train Loss: 0.0326 | Train Acc: 0.9884
Val Loss: 0.3749 | Val Acc: 0.9233

V5_GELU | Epoch [25/50]
Train Loss: 0.0322 | Train Acc: 0.9879
Val Loss: 0.3750 | Val Acc: 0.9258

V5_GELU | Epoch [26/50]
Train Loss: 0.0289 | Train Acc: 0.9894
Val Loss: 0.3694 | Val Acc: 0.9246

V5_GELU | Epoch [27/50]
Train Loss: 0.0320 | Train Acc: 0.9882
Val Loss: 0.3991 | Val Acc: 0.9231

V5_GELU | Epoch [28/50]
Train Loss: 0.0293 | Train Acc: 0.9889
Val Loss: 0.3819 | Val Acc: 0.9248

V5_GELU | Epoch [29/50]
Train Loss: 0.0263 | Train Acc: 0.9906
Val Loss: 0.4191 | Val Acc: 0.9248

V5_GELU | Epoch [30/50]
Train Loss: 0.0279 | Train Acc: 0.9902
Val Loss: 0.3994 | Val Acc: 0.9248

V5_GELU | Epoch [31/50]
Train Loss: 0.0239 | Train Acc: 0.9915
Val Loss: 0.4410 | Val Acc: 0.9259

V5_GELU | Epoch [32/50]
Train Loss: 0.0252 | Train Acc: 0.9909
Val Loss: 0.4047 | Val Acc: 0.9262

V5_GELU | Epoch [33/50]
Train Loss: 0.0263 | Train Acc: 0.9901
Val Loss: 0.4481 | Val Acc: 0.9227

V5_GELU | Epoch [34/50]
Train Loss: 0.0227 | Train Acc: 0.9913
Val Loss: 0.4386 | Val Acc: 0.9262

V5_GELU | Epoch [35/50]
Train Loss: 0.0228 | Train Acc: 0.9919
Val Loss: 0.4539 | Val Acc: 0.9224

V5_GELU | Epoch [36/50]
Train Loss: 0.0206 | Train Acc: 0.9924
Val Loss: 0.4747 | Val Acc: 0.9259

V5_GELU | Epoch [37/50]
Train Loss: 0.0233 | Train Acc: 0.9918
Val Loss: 0.4549 | Val Acc: 0.9222

V5_GELU | Epoch [38/50]
Train Loss: 0.0204 | Train Acc: 0.9923
Val Loss: 0.4750 | Val Acc: 0.9220

V5_GELU | Epoch [39/50]
Train Loss: 0.0208 | Train Acc: 0.9928
Val Loss: 0.4657 | Val Acc: 0.9228

V5_GELU | Epoch [40/50]
Train Loss: 0.0207 | Train Acc: 0.9924
Val Loss: 0.4859 | Val Acc: 0.9223

V5_GELU | Epoch [41/50]
Train Loss: 0.0231 | Train Acc: 0.9918
Val Loss: 0.4889 | Val Acc: 0.9234

V5_GELU | Epoch [42/50]
Train Loss: 0.0209 | Train Acc: 0.9927
Val Loss: 0.4978 | Val Acc: 0.9243

V5_GELU | Epoch [43/50]
Train Loss: 0.0148 | Train Acc: 0.9950
Val Loss: 0.5132 | Val Acc: 0.9269

V5_GELU | Epoch [44/50]
Train Loss: 0.0202 | Train Acc: 0.9925
Val Loss: 0.4895 | Val Acc: 0.9221

V5_GELU | Epoch [45/50]
Train Loss: 0.0219 | Train Acc: 0.9921
Val Loss: 0.5255 | Val Acc: 0.9236

V5_GELU | Epoch [46/50]
Train Loss: 0.0188 | Train Acc: 0.9936
Val Loss: 0.4992 | Val Acc: 0.9233

V5_GELU | Epoch [47/50]
Train Loss: 0.0175 | Train Acc: 0.9933
Val Loss: 0.5113 | Val Acc: 0.9252

V5_GELU | Epoch [48/50]
Train Loss: 0.0173 | Train Acc: 0.9938
Val Loss: 0.5559 | Val Acc: 0.9243

V5_GELU | Epoch [49/50]
Train Loss: 0.0168 | Train Acc: 0.9943
Val Loss: 0.5085 | Val Acc: 0.9229

V5_GELU | Epoch [50/50]
Train Loss: 0.0175 | Train Acc: 0.9936
Val Loss: 0.5274 | Val Acc: 0.9199

Training V6_NoDropout...
V6_NoDropout | Epoch [1/50]
Train Loss: 0.4982 | Train Acc: 0.8164
Val Loss: 0.3576 | Val Acc: 0.8710

V6_NoDropout | Epoch [2/50]
Train Loss: 0.3146 | Train Acc: 0.8869
Val Loss: 0.3154 | Val Acc: 0.8836

V6_NoDropout | Epoch [3/50]
Train Loss: 0.2694 | Train Acc: 0.9008

Val Loss: 0.2708 | Val Acc: 0.9014

V6_NoDropout | Epoch [4/50]

Train Loss: 0.2357 | Train Acc: 0.9144

Val Loss: 0.2594 | Val Acc: 0.9055

V6_NoDropout | Epoch [5/50]

Train Loss: 0.2102 | Train Acc: 0.9230

Val Loss: 0.2439 | Val Acc: 0.9106

V6_NoDropout | Epoch [6/50]

Train Loss: 0.1886 | Train Acc: 0.9297

Val Loss: 0.2417 | Val Acc: 0.9143

V6_NoDropout | Epoch [7/50]

Train Loss: 0.1671 | Train Acc: 0.9384

Val Loss: 0.2536 | Val Acc: 0.9082

V6_NoDropout | Epoch [8/50]

Train Loss: 0.1502 | Train Acc: 0.9443

Val Loss: 0.2572 | Val Acc: 0.9142

V6_NoDropout | Epoch [9/50]

Train Loss: 0.1319 | Train Acc: 0.9507

Val Loss: 0.2470 | Val Acc: 0.9144

V6_NoDropout | Epoch [10/50]

Train Loss: 0.1132 | Train Acc: 0.9573

Val Loss: 0.2538 | Val Acc: 0.9197

V6_NoDropout | Epoch [11/50]

Train Loss: 0.0997 | Train Acc: 0.9629

Val Loss: 0.2489 | Val Acc: 0.9181

V6_NoDropout | Epoch [12/50]

Train Loss: 0.0865 | Train Acc: 0.9679

Val Loss: 0.2678 | Val Acc: 0.9193

V6_NoDropout | Epoch [13/50]

Train Loss: 0.0742 | Train Acc: 0.9722

Val Loss: 0.2737 | Val Acc: 0.9195

V6_NoDropout | Epoch [14/50]

Train Loss: 0.0642 | Train Acc: 0.9767

Val Loss: 0.3232 | Val Acc: 0.9137

V6_NoDropout | Epoch [15/50]

Train Loss: 0.0552 | Train Acc: 0.9804

Val Loss: 0.3121 | Val Acc: 0.9189

V6_NoDropout | Epoch [16/50]

Train Loss: 0.0483 | Train Acc: 0.9829

Val Loss: 0.3556 | Val Acc: 0.9161

V6_NoDropout | Epoch [17/50]

Train Loss: 0.0397 | Train Acc: 0.9856

Val Loss: 0.3430 | Val Acc: 0.9197

V6_NoDropout | Epoch [18/50]

Train Loss: 0.0381 | Train Acc: 0.9861

Val Loss: 0.3663 | Val Acc: 0.9179

V6_NoDropout | Epoch [19/50]

Train Loss: 0.0329 | Train Acc: 0.9880

Val Loss: 0.3832 | Val Acc: 0.9198

V6_NoDropout | Epoch [20/50]

Train Loss: 0.0300 | Train Acc: 0.9893

Val Loss: 0.4098 | Val Acc: 0.9128

V6_NoDropout | Epoch [21/50]

Train Loss: 0.0255 | Train Acc: 0.9906

Val Loss: 0.4471 | Val Acc: 0.9173

V6_NoDropout | Epoch [22/50]

Train Loss: 0.0220 | Train Acc: 0.9927

Val Loss: 0.4509 | Val Acc: 0.9134

V6_NoDropout | Epoch [23/50]

Train Loss: 0.0236 | Train Acc: 0.9916

Val Loss: 0.4622 | Val Acc: 0.9170

V6_NoDropout | Epoch [24/50]

Train Loss: 0.0225 | Train Acc: 0.9922

Val Loss: 0.4775 | Val Acc: 0.9150

V6_NoDropout | Epoch [25/50]

Train Loss: 0.0171 | Train Acc: 0.9942

Val Loss: 0.4970 | Val Acc: 0.9182

V6_NoDropout | Epoch [26/50]

Train Loss: 0.0196 | Train Acc: 0.9933

Val Loss: 0.5104 | Val Acc: 0.9141

V6_NoDropout | Epoch [27/50]

Train Loss: 0.0239 | Train Acc: 0.9912

Val Loss: 0.6499 | Val Acc: 0.9038

V6_NoDropout | Epoch [28/50]

Train Loss: 0.0135 | Train Acc: 0.9954

Val Loss: 0.5578 | Val Acc: 0.9186

V6_NoDropout | Epoch [29/50]

Train Loss: 0.0178 | Train Acc: 0.9935

Val Loss: 0.5493 | Val Acc: 0.9143

V6_NoDropout | Epoch [30/50]

Train Loss: 0.0124 | Train Acc: 0.9957

Val Loss: 0.5491 | Val Acc: 0.9149

V6_NoDropout | Epoch [31/50]

Train Loss: 0.0190 | Train Acc: 0.9931

Val Loss: 0.5835 | Val Acc: 0.9163

V6_NoDropout | Epoch [32/50]

Train Loss: 0.0095 | Train Acc: 0.9973

Val Loss: 0.5445 | Val Acc: 0.9189

V6_NoDropout | Epoch [33/50]

Train Loss: 0.0161 | Train Acc: 0.9941

Val Loss: 0.5769 | Val Acc: 0.9110

V6_NoDropout | Epoch [34/50]

Train Loss: 0.0180 | Train Acc: 0.9936

Val Loss: 0.5902 | Val Acc: 0.9118

V6_NoDropout | Epoch [35/50]

Train Loss: 0.0100 | Train Acc: 0.9967

Val Loss: 0.6554 | Val Acc: 0.9175

V6_NoDropout | Epoch [36/50]

Train Loss: 0.0129 | Train Acc: 0.9954

Val Loss: 0.5905 | Val Acc: 0.9113

V6_NoDropout | Epoch [37/50]

Train Loss: 0.0160 | Train Acc: 0.9946

Val Loss: 0.6114 | Val Acc: 0.9188

V6_NoDropout | Epoch [38/50]

Train Loss: 0.0061 | Train Acc: 0.9979

Val Loss: 0.6296 | Val Acc: 0.9174

V6_NoDropout | Epoch [39/50]

Train Loss: 0.0170 | Train Acc: 0.9938

Val Loss: 0.6352 | Val Acc: 0.9164

V6_NoDropout | Epoch [40/50]

Train Loss: 0.0109 | Train Acc: 0.9961

Val Loss: 0.6226 | Val Acc: 0.9157

V6_NoDropout | Epoch [41/50]

Train Loss: 0.0143 | Train Acc: 0.9953

Val Loss: 0.6244 | Val Acc: 0.9173

V6_NoDropout | Epoch [42/50]

Train Loss: 0.0100 | Train Acc: 0.9964

Val Loss: 0.6660 | Val Acc: 0.9174

V6_NoDropout | Epoch [43/50]

Train Loss: 0.0139 | Train Acc: 0.9952

Val Loss: 0.6603 | Val Acc: 0.9176

V6_NoDropout | Epoch [44/50]

Train Loss: 0.0100 | Train Acc: 0.9966

Val Loss: 0.6666 | Val Acc: 0.9163

V6_NoDropout | Epoch [45/50]

Train Loss: 0.0082 | Train Acc: 0.9974

Val Loss: 0.7030 | Val Acc: 0.9120

V6_NoDropout | Epoch [46/50]

Train Loss: 0.0137 | Train Acc: 0.9952

Val Loss: 0.7172 | Val Acc: 0.9152

V6_NoDropout | Epoch [47/50]

Train Loss: 0.0097 | Train Acc: 0.9966

Val Loss: 0.6864 | Val Acc: 0.9183

V6_NoDropout | Epoch [48/50]

Train Loss: 0.0044 | Train Acc: 0.9987

Val Loss: 0.7193 | Val Acc: 0.9114

V6_NoDropout | Epoch [49/50]

Train Loss: 0.0151 | Train Acc: 0.9949

Val Loss: 0.7131 | Val Acc: 0.9187

V6_NoDropout | Epoch [50/50]

Train Loss: 0.0073 | Train Acc: 0.9972

Val Loss: 0.7321 | Val Acc: 0.9170

Training V7_Deeper...

V7_Deeper | Epoch [1/50]

Train Loss: 0.4670 | Train Acc: 0.8325

Val Loss: 0.3364 | Val Acc: 0.8731

V7_Deeper | Epoch [2/50]

Train Loss: 0.3093 | Train Acc: 0.8898

Val Loss: 0.2775 | Val Acc: 0.8952

V7_Deeper | Epoch [3/50]

Train Loss: 0.2681 | Train Acc: 0.9028

Val Loss: 0.2714 | Val Acc: 0.8982

V7_Deeper | Epoch [4/50]

Train Loss: 0.2390 | Train Acc: 0.9131

Val Loss: 0.2413 | Val Acc: 0.9126

V7_Deeper | Epoch [5/50]

Train Loss: 0.2213 | Train Acc: 0.9200

Val Loss: 0.2471 | Val Acc: 0.9112

V7_Deeper | Epoch [6/50]

Train Loss: 0.2030 | Train Acc: 0.9247

Val Loss: 0.2329 | Val Acc: 0.9155

V7_Deeper | Epoch [7/50]

Train Loss: 0.1879 | Train Acc: 0.9300

Val Loss: 0.2498 | Val Acc: 0.9100

V7_Deeper | Epoch [8/50]

Train Loss: 0.1701 | Train Acc: 0.9363

Val Loss: 0.2563 | Val Acc: 0.9101

V7_Deeper | Epoch [9/50]

Train Loss: 0.1580 | Train Acc: 0.9404

Val Loss: 0.2537 | Val Acc: 0.9123

V7_Deeper | Epoch [10/50]

Train Loss: 0.1473 | Train Acc: 0.9435

Val Loss: 0.2491 | Val Acc: 0.9184

V7_Deeper | Epoch [11/50]

Train Loss: 0.1386 | Train Acc: 0.9477

Val Loss: 0.2639 | Val Acc: 0.9166

V7_Deeper | Epoch [12/50]

Train Loss: 0.1295 | Train Acc: 0.9502

Val Loss: 0.2687 | Val Acc: 0.9103

V7_Deeper | Epoch [13/50]
Train Loss: 0.1214 | Train Acc: 0.9538
Val Loss: 0.2536 | Val Acc: 0.9191

V7_Deeper | Epoch [14/50]
Train Loss: 0.1135 | Train Acc: 0.9565
Val Loss: 0.3119 | Val Acc: 0.9070

V7_Deeper | Epoch [15/50]
Train Loss: 0.1052 | Train Acc: 0.9592
Val Loss: 0.2997 | Val Acc: 0.9089

V7_Deeper | Epoch [16/50]
Train Loss: 0.0948 | Train Acc: 0.9638
Val Loss: 0.2984 | Val Acc: 0.9109

V7_Deeper | Epoch [17/50]
Train Loss: 0.0915 | Train Acc: 0.9653
Val Loss: 0.2856 | Val Acc: 0.9167

V7_Deeper | Epoch [18/50]
Train Loss: 0.0874 | Train Acc: 0.9666
Val Loss: 0.3106 | Val Acc: 0.9135

V7_Deeper | Epoch [19/50]
Train Loss: 0.0826 | Train Acc: 0.9672
Val Loss: 0.3452 | Val Acc: 0.9079

V7_Deeper | Epoch [20/50]
Train Loss: 0.0813 | Train Acc: 0.9697
Val Loss: 0.3219 | Val Acc: 0.9147

V7_Deeper | Epoch [21/50]
Train Loss: 0.0759 | Train Acc: 0.9708
Val Loss: 0.3466 | Val Acc: 0.9146

V7_Deeper | Epoch [22/50]
Train Loss: 0.0690 | Train Acc: 0.9743
Val Loss: 0.3463 | Val Acc: 0.9137

V7_Deeper | Epoch [23/50]
Train Loss: 0.0662 | Train Acc: 0.9745
Val Loss: 0.3464 | Val Acc: 0.9167

V7_Deeper | Epoch [24/50]
Train Loss: 0.0658 | Train Acc: 0.9748
Val Loss: 0.3665 | Val Acc: 0.9139

V7_Deeper | Epoch [25/50]
Train Loss: 0.0592 | Train Acc: 0.9772
Val Loss: 0.4116 | Val Acc: 0.9115

V7_Deeper | Epoch [26/50]
Train Loss: 0.0597 | Train Acc: 0.9775
Val Loss: 0.3752 | Val Acc: 0.9121

V7_Deeper | Epoch [27/50]
Train Loss: 0.0549 | Train Acc: 0.9794
Val Loss: 0.4209 | Val Acc: 0.9085

V7_Deeper | Epoch [28/50]
Train Loss: 0.0556 | Train Acc: 0.9779
Val Loss: 0.3989 | Val Acc: 0.9126

V7_Deeper | Epoch [29/50]
Train Loss: 0.0559 | Train Acc: 0.9791
Val Loss: 0.3949 | Val Acc: 0.9144

V7_Deeper | Epoch [30/50]
Train Loss: 0.0509 | Train Acc: 0.9810
Val Loss: 0.4463 | Val Acc: 0.9125

V7_Deeper | Epoch [31/50]
Train Loss: 0.0507 | Train Acc: 0.9810
Val Loss: 0.4310 | Val Acc: 0.9123

V7_Deeper | Epoch [32/50]
Train Loss: 0.0464 | Train Acc: 0.9830
Val Loss: 0.4348 | Val Acc: 0.9095

V7_Deeper | Epoch [33/50]
Train Loss: 0.0449 | Train Acc: 0.9830
Val Loss: 0.4740 | Val Acc: 0.9133

V7_Deeper | Epoch [34/50]
Train Loss: 0.0465 | Train Acc: 0.9823
Val Loss: 0.4314 | Val Acc: 0.9143

V7_Deeper | Epoch [35/50]
Train Loss: 0.0435 | Train Acc: 0.9840
Val Loss: 0.5178 | Val Acc: 0.9094

V7_Deeper | Epoch [36/50]
Train Loss: 0.0431 | Train Acc: 0.9840
Val Loss: 0.4678 | Val Acc: 0.9148

V7_Deeper | Epoch [37/50]
Train Loss: 0.0408 | Train Acc: 0.9846
Val Loss: 0.4665 | Val Acc: 0.9132

V7_Deeper | Epoch [38/50]
Train Loss: 0.0373 | Train Acc: 0.9859
Val Loss: 0.4515 | Val Acc: 0.9132

V7_Deeper | Epoch [39/50]
Train Loss: 0.0407 | Train Acc: 0.9841
Val Loss: 0.4841 | Val Acc: 0.9140

V7_Deeper | Epoch [40/50]
Train Loss: 0.0381 | Train Acc: 0.9858
Val Loss: 0.4726 | Val Acc: 0.9146

V7_Deeper | Epoch [41/50]
Train Loss: 0.0381 | Train Acc: 0.9857
Val Loss: 0.4916 | Val Acc: 0.9119

V7_Deeper | Epoch [42/50]
Train Loss: 0.0370 | Train Acc: 0.9864
Val Loss: 0.5212 | Val Acc: 0.9099

V7_Deeper | Epoch [43/50]
 Train Loss: 0.0377 | Train Acc: 0.9863
 Val Loss: 0.4924 | Val Acc: 0.9058

V7_Deeper | Epoch [44/50]
 Train Loss: 0.0370 | Train Acc: 0.9863
 Val Loss: 0.5119 | Val Acc: 0.9128

V7_Deeper | Epoch [45/50]
 Train Loss: 0.0345 | Train Acc: 0.9871
 Val Loss: 0.5026 | Val Acc: 0.9103

V7_Deeper | Epoch [46/50]
 Train Loss: 0.0328 | Train Acc: 0.9878
 Val Loss: 0.5251 | Val Acc: 0.9130

V7_Deeper | Epoch [47/50]
 Train Loss: 0.0341 | Train Acc: 0.9876
 Val Loss: 0.5528 | Val Acc: 0.9128

V7_Deeper | Epoch [48/50]
 Train Loss: 0.0337 | Train Acc: 0.9877
 Val Loss: 0.5543 | Val Acc: 0.9119

V7_Deeper | Epoch [49/50]
 Train Loss: 0.0326 | Train Acc: 0.9883
 Val Loss: 0.5270 | Val Acc: 0.9122

V7_Deeper | Epoch [50/50]
 Train Loss: 0.0290 | Train Acc: 0.9890
 Val Loss: 0.6293 | Val Acc: 0.9073

All models completed.

Sorted by best validation accuracy:

	model	best_val_accuracy	best_val_epoch	best_val_loss \
0	V1_Baseline	0.926917	22	0.215565
1	V2_BatchNorm	0.926083	17	0.214957
2	V3_LargerKernel	0.924500	46	0.226009
3	V4_AvgPool	0.926333	50	0.235536
4	V5_GELU	0.926917	43	0.233052
5	V6_NoDropout	0.919833	19	0.241726
6	V7_Deeper	0.919083	13	0.232922

	best_val_loss_epoch	final_train_loss	final_train_accuracy \
0	8	0.030818	0.987938
1	8	0.027186	0.990125
2	9	0.032323	0.988000
3	16	0.057241	0.977271
4	6	0.017539	0.993583
5	6	0.007290	0.997188
6	6	0.029007	0.988979

	final_val_loss	final_val_accuracy
0	0.534767	0.920083
1	0.498857	0.922917
2	0.565055	0.920917
3	0.376946	0.926333
4	0.527369	0.919917
5	0.732076	0.917000

6 0.629253 0.907333

Sorted by best validation loss:

	model	best_val_accuracy	best_val_epoch	best_val_loss \
1	V2_BatchNorm	0.926083	17	0.214957
0	V1_Baseline	0.926917	22	0.215565
2	V3_LargerKernel	0.924500	46	0.226009
6	V7_Deeper	0.919083	13	0.232922
4	V5_GELU	0.926917	43	0.233052
3	V4_AvgPool	0.926333	50	0.235536
5	V6_NoDropout	0.919833	19	0.241726

	best_val_loss_epoch	final_train_loss	final_train_accuracy \
1	8	0.027186	0.990125
0	8	0.030818	0.987938
2	9	0.032323	0.988000
6	6	0.029007	0.988979
4	6	0.017539	0.993583
3	16	0.057241	0.977271
5	6	0.007290	0.997188

	final_val_loss	final_val_accuracy
1	0.498857	0.922917
0	0.534767	0.920083
2	0.565055	0.920917
6	0.629253	0.907333
4	0.527369	0.919917
3	0.376946	0.926333
5	0.732076	0.917000